



**JOMO KENYATTA UNIVERSITY OF
AGRICULTURE & TECHNOLOGY
COMPUTING DEPARTMENT**

ICS 2313: SOFTWARE TESTING & QUALITY ASSURANCE

APPLICATION TITLE: COHORT SPACE

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1. Problem Statement

Modern student group projects are increasingly undermined by fragmented digital workflows. Team members rely on general-purpose messaging platforms such as WhatsApp or Telegram to coordinate their work, which results in important resources — including links, documents, and task assignments — being buried in lengthy chat histories and eventually lost.

Furthermore, task ownership within these groups is often ambiguous; multiple members may attempt the same task while others remain idle, a phenomenon commonly referred to as social loafing. Academic milestones imposed by instructors are also frequently decoupled from the team's day-to-day progress tracking, leading to a disconnect between what needs to be done and what is actually being done.

The cumulative effect of these issues is a pattern of last-minute submission chaos, where teams scramble to consolidate work at the deadline, resulting in poor quality deliverables, unequal contribution, and unnecessary academic stress. There is a clear need for a purpose-built platform that addresses these challenges holistically within an academic context.

Proposed Solution

The proposed solution is an Agile Academic Management Hub — a web-based application that centralizes the entire project lifecycle for student teams. The system integrates a persistent resource vault for storing links and documents alongside a milestone-driven task tracker. It provides teams with a single, PIN-protected workspace to visualize progress, enforce individual accountability, and synchronize project deliverables with semester deadlines.

2. Requirements

2.1 Functional Requirements

The following functional requirements define the core capabilities the system must provide:

ID	Category	Description
FR-01	Authentication	The system shall allow teams to create and access a PIN-protected workspace accessible only to authorized members.
FR-02	Resource Vault	The system shall provide a persistent vault where members can upload, store, retrieve, and share project documents and links.
FR-03	Task Management	The system shall allow members to create, assign, and track tasks with deadlines, ownership, and real-time status updates (To Do, In Progress, Done).
FR-04	Milestone Tracking	The system shall allow teams to define academic milestones, map them to semester deadlines, and visualize progress via a timeline or progress bar.
FR-05	Dashboard	The system shall provide a centralized dashboard aggregating team task completion rates and overall milestone status at a glance.
FR-06	Accountability	The system shall display per-member contribution metrics showing individual task completion to identify and reduce social loafing.

2.2 Non-functional Requirements

The following non-functional requirements define the core quality attributes the system must provide:

ID	Category	Description
NR-01	Performance	The system shall support simultaneous access by multiple team members without noticeable delay. Changes and updates should be real time.
NR-02	Usability	The system shall be easy to use. The UI shall be simple and intuitive and use consistent pattern in terms of design
NR-03	Reliability	The system shall be reliable, task updates and resources to persist after restarts or logouts.
NR-04	Availability	The system shall allow teams to use anytime, 24/7 with minimal downtime. Maintenance to occur low usage periods.
NR-05	Scalability	The system shall be scalable as the usages goes up. The database and the server architecture shall allow scalability to occupy more users and projects
NR-06	Compatibility	The system shall work on any browser, Chrome, Firefox e.t.c across different environments

2.3 Hardware and Software Tools

Hardware Tools

- **Developer Laptop/Personal Computer:** For developers to write code, run environments, and test the application.
- **Internet Connectivity:** Stable access required for cloud services, version control, online documentation, and team collaboration.
- **Mobile Devices (Smartphones/Tablets):** Used to test the application's responsiveness and usability on different screen sizes for accessibility.
- **Cloud Server Infrastructure:** Used for deploying and running the Cohort Space platform, enabling remote access and real-time collaboration.

Software Tools

- **Backend Framework – Node.js:** Implements server-side logic, managing multi-user access, APIs, authentication, and database integration.
- **Database Management System – PostgreSQL:** Stores structured data such as user accounts, project milestones, tasks, and resources.
- **Design Tool – Figma:** Used for UI/UX design, creating wireframes and prototypes collaboratively.
- **Frontend Framework – React.js:** Builds the interactive and responsive user interface of the platform.
- **Project Management Tool – GitHub:** Manages version control, collaborative coding, and tracks project progress.
- **Testing Tool – Selenium:** Used for automated testing to ensure the web application functions correctly across different browsers.
- **Web Browser – Google Chrome:** Used for development and testing due to its robust debugging and performance monitoring tools.

PROJECT PLAN

NO	ACTIVITY	START DATE	COMPLETION DATE	DELIVERABLE
1	Proposal Writing	03/03/2026	09/03/2026	Project Proposal
2	Design	10/03/2026	23/03/2026	Design Document
3	Coding	24/03/2026	20/04/2026	Source Code
4	Testing	21/04/2026	4/05/2026	Test Results
5	Documentation	5/05/2026	11/05/2026	User Manual
6	Go Live	12/05/2026	18/05/2026	Pre-Go Live & Post Go Live Results